Erased or Hidden? Unlearned chemistry in Tiny Aya returns after fine-tuning in any language, and how much returns depends on how much was erased, not where

Janhavi Khindkar · ValueShift Research · results report, 13 November 2026 · model: Tiny Aya Global (3.35B)

Summary

- **Unlearned knowledge is hidden, not erased.** After English-only RMU unlearning of high-school chemistry, a small LoRA fine-tune brings held-out answers back in every arm. Pooled over both test halves: layer 27 +0.27 (95% CI +0.12 to +0.42); layer 27, deeper +0.43 (95% CI +0.27 to +0.59); layer 30 +0.33 (95% CI +0.19 to +0.47).
- **Any language, any topic.** Fine-tuning in Hindi or Marathi restores *English* answers as much as fine-tuning in English (layer 27, deeper: +0.53 English, +0.52 Hindi, +0.56 Marathi). A fine-tune on **unrelated** Marathi general-knowledge questions, with no chemistry at all, restores the most (+0.69 and +0.62; even the shallow layer-27 arm +0.30, 95% CI +0.11 to +0.50).
- **Depth, not layer.** Across 4 arms × 3 languages, what comes back tracks how much was erased: relearning excess ≈ 0.57 × erased +0.09 ($r = 0.86$); layer 27 and 30 sit -0.02 and +0.02 off that line. At matched English depth (layer 27 deeper vs layer 30) the English difference is -0.03 (95% CI -0.08 to +0.03).
- **The original hypothesis is not supported.** We expected unlearning at the more cross-lingually shared layer to be more durable. Its apparent advantage in the first runs came from erasing less. Durability comparisons between unlearning settings need matched erasure depth, per language.

1. Question and design

Question. When unlearning makes a model fail questions on a topic, is the knowledge gone, or only hidden? Does it come back when someone fine-tunes the model in another language, or on unrelated data, and does that depend on which layer the unlearning edited?

Model and data. Tiny Aya Global (36 layers). Forget topic: MMLU high-school chemistry, in English, Hindi (MMMLU) and Marathi (machine-translated with NLLB-200). The **forget set** is the 59 questions the model answers correctly in both English and Hindi; the **control set** is the 105 it gets wrong in both. Each is split into halves A and B: attacks train on one half and are tested on the other, in both directions for the main arms. Collateral damage is measured on college chemistry (ring 1), high-school biology and physics (ring 2), prehistory and international law (ring 3), and 200 general MMLU questions.

Unlearning. RMU (Li et al., WMDP): activations at layer ℓ on English forget text are pushed towards a random direction while activations on Wikipedia text are held fixed; only the MLP down-projections of layers $\ell-2 \ldots \ell$ are trained, on English text only. Arms: A2 (layer 27), A2s (layer 27, trained longer to erase as deeply as layer 30), A1c and A1 (layer 30, medium and strong), plus two gentle layer-30 settings that only half-erased.

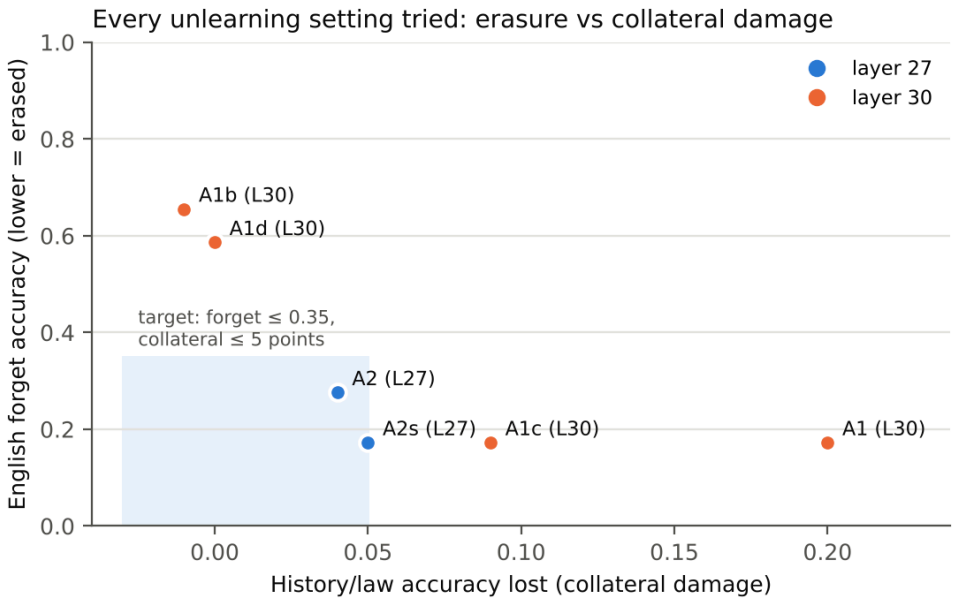
Attacks. LoRA (rank 16, all linear layers, 3 epochs), learning rates 1e-4 and 3e-4, 2 seeds; the strongest learning rate per seed is kept. Targeted attacks train on forget questions with their answers, in English, Hindi or Marathi (29–30 questions). Benign attacks train in Marathi on 100 college-chemistry questions, or on 200 general-knowledge questions from unrelated subjects. **Relearning excess** = accuracy gain on held-out forget questions minus the gain the same attack gives on held-out control questions (questions the model never knew), so a model that only gets better at the multiple-choice format scores zero.

Scoring. Multiple choice, by comparing the model's scores for the letters A–D after the question (no generation, no judge). The model also writes free-text answers for 8 held-out questions per language under each condition.

2. Erasure and collateral damage

Arm	Forget en	Forget hi	Forget mr	History/law en	Bio/physics en	General en
A0 original model	1.00	1.00	0.57	0.61	0.40	0.40
A2 layer 27	0.50	0.53	0.37	0.57	0.31	0.40
A2s layer 27, deeper	0.20	0.30	0.27	0.56	0.21	0.36
A1c layer 30, medium	0.23	0.53	0.30	0.52	0.17	0.35
A1 layer 30, strong	0.20	0.37	0.30	0.41	0.16	0.32

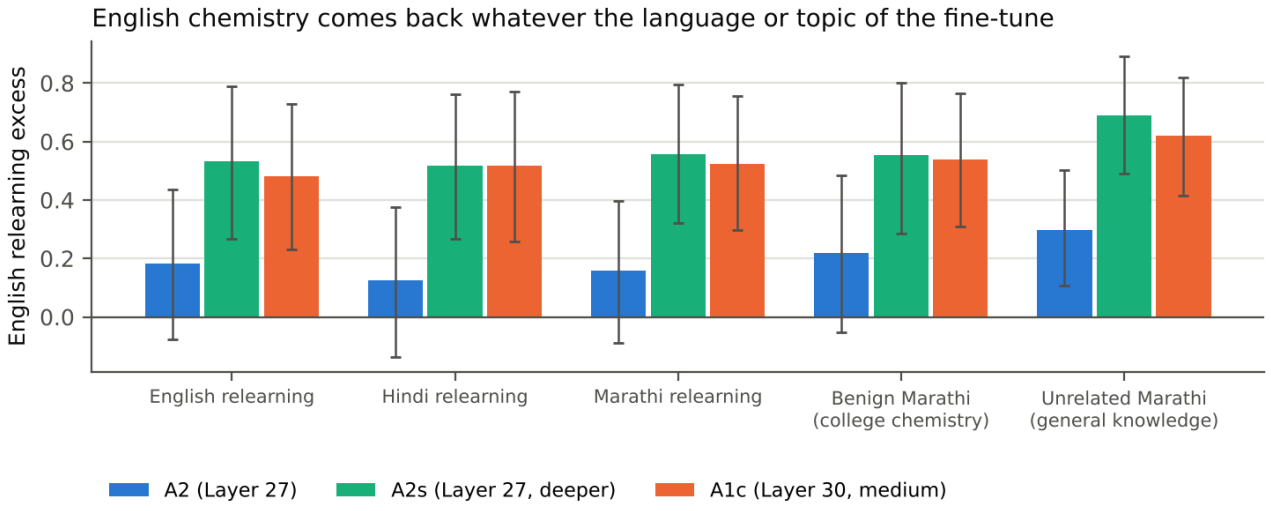
Accuracy on held-out half B before any relearning. The original model scores 1.00 / 1.00 / 0.57 on the forget set by construction.



Arm	Layer	Steps	Learning rate	Strength	Forget en (half A)	History/law drop	Matched
A2	27	150	5e-05	2	0.28	+4 pts	yes
A2s	27	300	5e-05	2	0.17	+5 pts	yes
A1	30	400	0.0001	4	0.17	+20 pts	no
A1b	30	400	5e-05	2	0.66	-1 pts	no
A1c	30	200	0.0001	4	0.17	+9 pts	no
A1d	30	250	5e-05	2	0.59	+0 pts	no

English-only unlearning also erased the knowledge in Hindi and Marathi at both layers. Layer 27 gave a moderate setting (A2) that met the target first time; the layer-30 settings tried jumped from half-erased (0.59–0.66) to fully erased, so we cannot say whether a moderate layer-30 setting exists. **At matched depth the two layers damage about the same:** A2s and A1c both erase English to 0.17 and leave biology/physics at 0.21 and 0.17, general questions at 0.36 and 0.35.

3. The knowledge comes back, in any language and from any topic



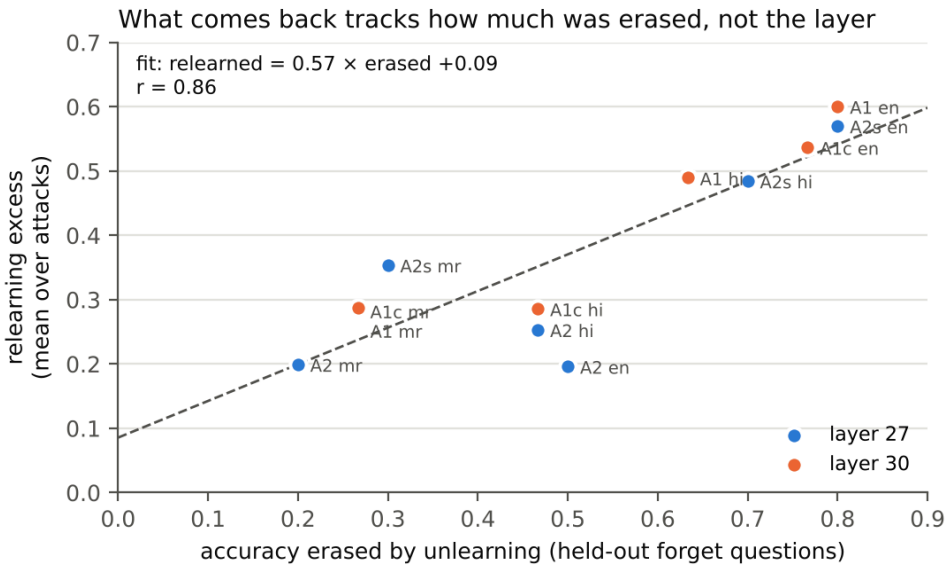
English relearning excess (95% CI over questions) after each attack, for the three main arms (half B).

Attack	Scored in	**A2** layer 27	**A2s** layer 27, deeper	**A1c** layer 30, medium	**A1** layer 30, strong
English relearning	English	+0.18 [-0.08, +0.44]	+0.53 [+0.27, +0.79]	+0.48 [+0.23, +0.73]	+0.61 [+0.36, +0.84]
English relearning	Hindi	+0.21 [-0.05, +0.47]	+0.46 [+0.19, +0.73]	+0.18 [-0.11, +0.46]	+0.41 [+0.15, +0.67]
English relearning	Marathi	+0.22 [-0.02, +0.46]	+0.33 [+0.06, +0.58]	+0.31 [+0.05, +0.57]	+0.27 [-0.02, +0.54]
Hindi relearning	English	+0.13 [-0.14, +0.38]	+0.52 [+0.27, +0.76]	+0.52 [+0.26, +0.77]	+0.63 [+0.38, +0.86]
Hindi relearning	Hindi	+0.27 [+0.01, +0.54]	+0.46 [+0.17, +0.73]	+0.32 [+0.04, +0.60]	+0.51 [+0.24, +0.77]
Hindi relearning	Marathi	+0.19 [-0.07, +0.46]	+0.39 [+0.13, +0.65]	+0.35 [+0.10, +0.61]	+0.34 [+0.07, +0.60]
Marathi relearning	English	+0.16 [-0.09, +0.40]	+0.56 [+0.32, +0.79]	+0.52 [+0.30, +0.76]	+0.59 [+0.34, +0.82]
Marathi relearning	Hindi	+0.29 [+0.03, +0.55]	+0.47 [+0.21, +0.72]	+0.32 [+0.06, +0.59]	+0.52 [+0.26, +0.77]
Marathi relearning	Marathi	+0.17 [-0.11, +0.44]	+0.32 [+0.06, +0.59]	+0.22 [-0.04, +0.49]	+0.29 [+0.03, +0.57]
Benign Marathi (college chemistry)	English	+0.22 [-0.05, +0.48]	+0.55 [+0.29, +0.80]	+0.54 [+0.31, +0.76]	+0.55 [+0.31, +0.77]
Benign Marathi (college chemistry)	Hindi	+0.20 [-0.09, +0.48]	+0.45 [+0.17, +0.72]	+0.23 [-0.02, +0.47]	+0.45 [+0.19, +0.69]
Benign Marathi (college chemistry)	Marathi	+0.15 [-0.14, +0.42]	+0.27 [+0.01, +0.53]	+0.21 [-0.04, +0.45]	+0.17 [-0.10, +0.45]
Unrelated Marathi (general knowledge)	English	+0.30 [+0.11, +0.50]	+0.69 [+0.49, +0.89]	+0.62 [+0.41, +0.82]	+0.64 [+0.43, +0.83]
Unrelated Marathi (general knowledge)	Hindi	+0.29 [+0.06, +0.53]	+0.58 [+0.33, +0.81]	+0.38 [+0.12, +0.63]	+0.56 [+0.37, +0.75]
Unrelated Marathi (general knowledge)	Marathi	+0.27 [+0.03, +0.53]	+0.46 [+0.21, +0.71]	+0.32 [+0.10, +0.53]	+0.36 [+0.13, +0.59]

Arm	Questions	Relearning excess	P(≤ 0)
A2 layer 27	both halves (59)	+0.27 [+0.12, +0.42]	0.001
A2s layer 27, deeper	both halves (59)	+0.43 [+0.27, +0.59]	0.000
A1c layer 30, medium	both halves (59)	+0.33 [+0.19, +0.47]	0.000
A1 layer 30, strong	half B (30)	+0.43 [+0.21, +0.64]	0.000

Every arm recovers held-out answers, and neither the attack language nor its topic matters much. The unrelated Marathi fine-tune is the strongest attack, but it also trains on the most questions (200, against 29–30 for targeted attacks), so attack types should not be ranked against each other. What it shows is that **no forgotten question, no chemistry and no English is needed** to bring English chemistry back.

4. Depth, not layer



Each point is one arm scored in one language: accuracy erased by unlearning (x) against relearning excess averaged over the five attacks (y).

Comparison (same questions)	Folds	Scored in	Difference in relearning excess	P(≤ 0)
A1c − A2s	1 + 2	all three	-0.10 [-0.16, -0.03]	1.000
A1c − A2s	1 + 2	English	-0.03 [-0.08, +0.03]	0.825
A1c − A2s	1 + 2	Hindi	-0.23 [-0.36, -0.10]	1.000
A1c − A2s	1 + 2	Marathi	-0.04 [-0.12, +0.03]	0.866
A1c − A2	1 + 2	all three	+0.06 [-0.03, +0.15]	0.103
A1c − A2	1 + 2	English	+0.22 [+0.10, +0.34]	0.000
A1c − A2	1 + 2	Hindi	-0.09 [-0.24, +0.05]	0.891
A1c − A2	1 + 2	Marathi	+0.04 [-0.05, +0.15]	0.192

In the first runs, layer 30 recovered more than layer 27 (A1c − A2), which looked like support for the hypothesis that unlearning at the shared layer is more durable. But A1c had erased much more English, and the difference sat entirely in English. Erasing as deeply at layer 27 (A2s) removes it: at matched English depth the English difference is -0.03, and overall layer 27 recovers slightly *more* (-0.10), driven by Hindi, where A2s had erased more (0.30 vs 0.53). The scatter shows the same thing across all twelve arm × language points: how much comes back is roughly 0.57 of how much was erased, whichever layer was edited.

5. What the model actually says

Unlearned models do not give wrong answers; they produce degenerate text. After a Marathi fine-tune they answer fluently again, on questions they were never trained on. Greedy decoding, prompt: “give the letter, then explain in one sentence”.

Monatomic ions of the representative elements are often

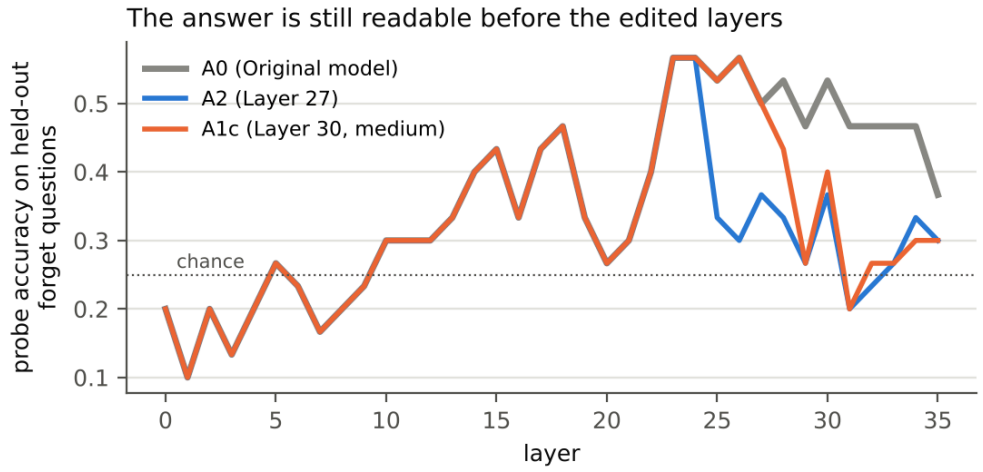
Correct: C. isoelectronic with a noble gas

A0	✓	`C. isoelectronic with a noble gas Monatomic ions of the representative elements are often isoelectronic with a noble gas because they have the same n…`
A1c	×	`"0 " (0 " **0** " " "

Condition	English	Marathi
A0	5/8	4/8
A1	0/8	0/8
A1 + Marathi attack relearning	2/8	3/8
A1 + benign Marathi relearning	3/8	4/8
A2	1/8	1/8
A2 + Marathi attack relearning	3/8	4/8
A2 + benign Marathi relearning	3/8	4/8
A1c	0/8	0/8
A1c + Marathi attack relearning	3/8	3/8
A1c + benign Marathi relearning	2/8	4/8

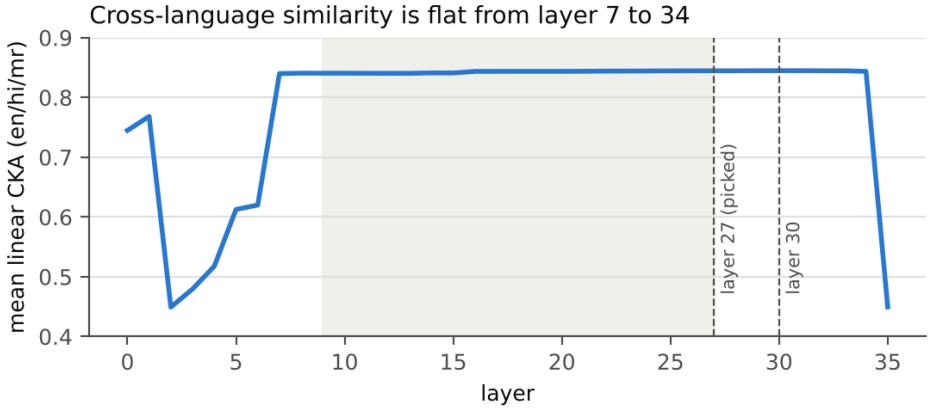
Correct letter given first, on 8 held-out questions per language. The original model itself gets 5/8 in this free-text format. Eight questions illustrate; they do not test.

6. Probe: the answer is readable before the edited layers



A linear probe trained on the original model reads the correct option on held-out forget questions at about 0.6 at layers 23–24 (chance 0.25), identically for every arm, because RMU leaves everything before the edited layers untouched; accuracy falls to chance only after the edit. RMU blocks the read-out of knowledge that is still represented upstream, which is what relearning undoes. **Caveat:** the probe's cross-validated accuracy on its mixed training set is near chance (0.27), the readout uses 30 questions, and the layers were picked after looking. Supporting evidence only.

7. A correction to the layer choice



Layer 27 was chosen as the layer where English, Hindi and Marathi representations are most similar (linear CKA). The curve is flat, 0.840–0.845 at every layer from 7 to 34, and 27 was simply the upper edge of the search window. The comparison is **layer 27 vs layer 30**, two late depths three layers apart. A better measure of cross-lingual sharing, and a genuinely middle layer, are the obvious next test of the original hypothesis.

8. What this means

- **For anyone relying on unlearning in a multilingual model:** RMU-style unlearning can be undone by fine-tuning in any language the model knows, including on data unrelated to the forgotten topic. Evaluations that only re-attack in English, or only with forget data, underestimate how easily it comes back.
- **For comparing unlearning settings or methods:** compare durability at matched erasure depth, per language. A shallower setting will look more durable simply because it has less to give back.

9. Limitations

- 59 forget questions; one model; one unlearning method (RMU); two seeds that vary only the attack.
- Both halves were tested only for A2, A2s and A1c; A1 has half B only.
- Attack types differ in size (29–30, 100 and 200 training questions), so their strengths are not comparable with each other.
- Marathi is machine-translated, and the model is near chance in Marathi chemistry (0.32 overall).
- Chemistry formulas and numbers look the same in every language, so some transfer may run through shared symbols.
- The cross-language similarity measure did not single out any layer, and only two late layers were compared.

10. Related work

Li et al., *The WMDP Benchmark* (RMU). Łucki et al., *An Adversarial Perspective on Machine Unlearning for AI Safety*, and Hu et al., *Unlearning or Obfuscating?*: relearning and unrelated fine-tuning restore unlearned knowledge, in one language. Deeb & Roger, *Do Unlearning Methods Remove Information from Language Model Weights?*. *Linguistic Loopholes in LLM Unlearning* (2609.40286): Tiny Aya and RMU across 174 languages, changing only the test-time query language, with no relearning or layer analysis. *Multilingual Unlearning in LLMs: Transfer, Dynamics, and Reversibility* (2606.03291): reversal by activation steering, not fine-tuning. *A False Positive in Cross-Lingual Unlearning Evaluation*: cross-lingual claims need base-model and retain controls, which this design includes (A0, control set, rings).

11. Reproduce

```
RESULTS_REPO=yourname/eoh-results bash run_lambda.sh      # pipeline (or the Colab notebook / HF Jobs)
RESULTS_REPO=yourname/eoh-results bash run_variants.sh    # layer-30 settings + generated samples
RESULTS_REPO=yourname/eoh-results bash run_robust.sh      # A2s, Hindi + unrelated attacks, swapped halves
python analysis/analyze.py --results results --out analysis && python report/build_report.py
```